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Bone Strength Analyzer Using Multimodal Sensors and Hybrid CNN–LSTM Model

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Abstract-Bone strength degradation is a silent and progressive condition that significantly increases the risk of fractures, osteoporosis, and long-term mobility impairment. Conventional diagnostic techniques such as dual-energy X-ray absorptiometry (DEXA) provide reliable bone mineral density measurements; however, they are expensive, radiation-based, clinic-dependent, and unsuitable for continuous monitoring. This paper presents a portable, non-invasive bone strength analysis system that integrates multimodal sensor data with a hybrid deep learning model for qualitative bone health assessment. The proposed system combines inertial motion sensing, pressure response measurement, and optional bone imaging to capture complementary mechanical and structural characteristics of bone tissue. Motion and vibration signals are acquired using an MPU6050 inertial sensor, while force response data are obtained through a piezoelectric transducer. Bone surface or radiographic images provide additional spatial information. A hybrid Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) model is employed to extract spatial and temporal features, respectively. Feature-level fusion enables robust classification of bone strength into clinically meaningful categories. The system emphasizes portability, affordability, and radiation-free operation, making it suitable for continuous bone health monitoring in both clinical and personal healthcare environments.

Keywords— Bone strength assessment, multimodal sensing, MPU6050, piezoelectric sensor, CNN–LSTM, deep learning, embedded healthcare systems.

I. Introduction

Bone health plays a critical role in maintaining structural stability, mobility, and overall quality of life. Progressive bone weakening caused by aging, nutritional deficiencies, hormonal imbalance, and sedentary lifestyles often remains asymptomatic until fractures occur. Osteoporosis alone affects millions of individuals worldwide and is a major contributor to disability among the elderly population. Early detection and continuous monitoring of bone strength are therefore essential for preventive healthcare and timely medical intervention.

Current bone assessment techniques rely heavily on imaging-based modalities such as dual-energy X-ray absorptiometry (DEXA), quantitative computed tomography, and ultrasound-based methods. Although these techniques provide clinically validated results, they suffer from several limitations. Most systems are expensive, require specialized infrastructure, expose patients to radiation, and cannot be deployed for frequent or home-based monitoring. These constraints restrict their accessibility, particularly in rural and resource-limited regions.

Recent advancements in embedded systems, wearable sensors, and artificial intelligence have opened new possibilities for non-invasive health monitoring solutions. Sensor-based techniques can capture mechanical responses of biological tissues, while data-driven models can learn complex relationships between physiological signals and health conditions. Multimodal sensing, which combines heterogeneous data sources, has been shown to improve reliability and robustness compared to single-sensor approaches.

This work proposes a smart Bone Strength Analyzer that integrates motion, vibration, pressure, and imaging data with a hybrid deep learning framework. The system is designed to be portable, low-cost, and suitable for continuous monitoring. By fusing sensor signals with image-based features, the proposed approach provides a qualitative assessment of bone strength without relying on radiation-based diagnostics.

II. Related Work

Bone health assessment has been widely studied using imaging, signal processing, and machine learning techniques. Traditional approaches primarily focus on bone mineral density estimation using DEXA scans, which are considered the clinical gold standard. However, these methods are limited by cost, radiation exposure, and restricted availability.

Non-invasive alternatives based on vibration and acoustic signal analysis have been explored to assess joint and bone conditions. Vibroarthographic signal analysis techniques capture surface vibrations generated during joint movement and have demonstrated potential for early detection of bone and joint abnormalities. While promising, these methods often rely on handcrafted features and lack generalization across diverse populations.

With the rise of deep learning, convolutional neural networks have been extensively applied to bone image analysis, particularly for osteoporosis detection and fracture identification. CNN-based models achieve high accuracy in extracting spatial features from radiographs and computed tomography images. However, image-only approaches fail to capture dynamic mechanical properties of bones and require large labeled datasets.

Recurrent neural networks, especially Long Short-Term Memory models, have been used for analyzing biomedical time-series data such as gait signals, vibration patterns, and physiological measurements. LSTM networks effectively model temporal dependencies but are insufficient when spatial or structural information is required.

Recent studies emphasize the importance of multimodal data fusion in medical decision-support systems. By combining sensor signals and image data, multimodal frameworks improve diagnostic robustness and reduce uncertainty. The proposed system extends this concept by integrating low-cost embedded sensors with a hybrid CNN–LSTM architecture for bone strength assessment in non-clinical environments.

III. Proposed System Architecture

The proposed Bone Strength Analyzer is designed using a layered architecture to ensure modularity, scalability, and efficient integration of heterogeneous data sources. The system is organized into three primary layers: sensing, processing, and analysis. This structured design enables independent development and optimization of each layer while maintaining seamless data flow across the overall system.

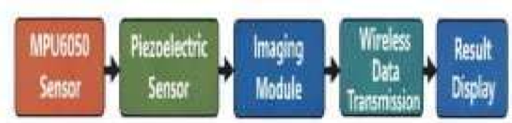
The sensing layer is responsible for acquiring multimodal information related to bone behavior. Motion and vibration signals are captured using the MPU6050 inertial measurement unit, which provides three-axis accelerometer and gyroscope data. These signals reflect the dynamic response of bone tissue during controlled movement or excitation. In parallel, a piezoelectric transducer is employed to measure pressure and force response, converting mechanical stress into electrical signals that represent localized mechanical resistance of the bone. An imaging module is incorporated to obtain bone surface or radiographic images, providing complementary structural and spatial information that cannot be derived from sensor signals alone.

The processing layer centers on the NodeMCU microcontroller, which serves as the system's control and communication unit. The NodeMCU interfaces with all sensing components and manages synchronized data acquisition across multiple modalities. Preliminary signal conditioning, including analog-to-digital conversion and basic filtering, is performed at this stage to ensure signal integrity. The integrated wireless communication capability of the NodeMCU enables real-time transmission of acquired data to an external processing unit or cloud-based platform for further analysis.

The analysis layer implements advanced data processing and intelligent decision-making functions. Preprocessed sensor signals and image data are forwarded to the deep learning module, where feature extraction and fusion are performed. The hybrid CNN–LSTM model operates within this layer, extracting spatial features from image data and temporal patterns from sequential sensor signals. Feature-level fusion combines these representations into a unified descriptor, which is subsequently used for qualitative bone strength classification.

The modular nature of the proposed architecture allows future expansion without major structural modifications. Additional sensors, enhanced communication protocols, or updated analytical models can be integrated seamlessly. This flexibility makes the system adaptable to evolving healthcare requirements and supports long-term deployment in both clinical screening and personal health monitoring applications.

Fig. 1. Block diagram of the proposed Bone Strength Analyzer



(a) Fig. 1. Block diagram of the proposed Bone Strength Analyzer.

IV. Methodology

A. Data Acquisition

Data acquisition is conducted under controlled experimental conditions to ensure repeatability and consistency. Subjects are instructed to perform predefined movements or maintain specific postures while sensor data are recorded. Imaging data are captured using a fixed camera orientation to minimize variability.

B. Signal Preprocessing

Raw sensor signals are prone to noise caused by environmental interference and sensor drift. Digital filtering techniques, including low-pass and band-pass filters, are applied to isolate relevant frequency components. Signal normalization is performed to standardize amplitude ranges across samples.

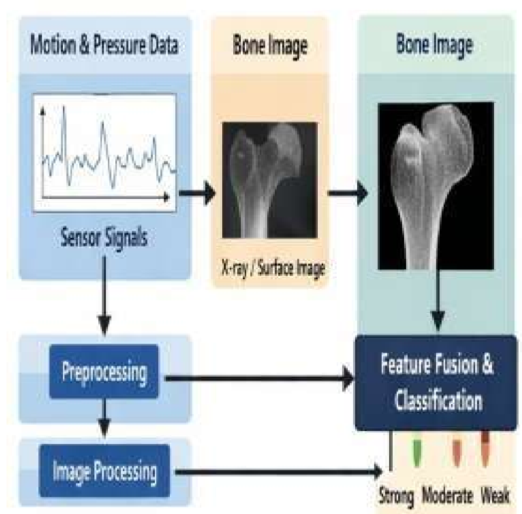
C. Image Preprocessing

Captured images are resized to standardized dimensions compatible with the CNN input. Intensity normalization and optional data augmentation techniques, such as rotation and scaling, are employed to enhance model generalization.

D. Multimodal Data Synchronization

Temporal alignment of sensor signals and image frames is achieved using timestamp-based synchronization. This ensures that multimodal data segments correspond to the same physical event, enabling effective feature-level fusion.

Fig. 2. System workflow of sensor data and image fusion



(b) Fig. 2. System workflow of sensor data and image fusion.

V. Hardware Implementation

The hardware prototype is designed with an emphasis on portability, low power consumption, and ease of deployment. The NodeMCU serves as the central controller, managing sensor communication and wireless data transmission.

The MPU6050 provides high-resolution three-axis accelerometer and gyroscope measurements, enabling detection of subtle vibration and motion patterns. The piezoelectric sensor converts mechanical stress into electrical signals, offering high sensitivity to pressure variations. The imaging module supplies structural information that complements sensor-based measurements.

An OLED display is incorporated for real-time visualization of qualitative bone strength assessment results.

TableI
Hardware Components and Specifications

Component	Specification	Function
NodeMCU (ESP8266)	Wi-Fi enabled MCU	Data processing & transmission
MPU6050	3-axis accelerometer & gyroscope	Motion and vibration sensing
Piezoelectric Sensor	High sensitivity	Pressure response detection
Imaging Module	CMOS camera	Bone image acquisition
OLED Display	128×64	Result visualization

The NodeMCU acts as the central controller, interfacing all sensors and managing communication with the processing unit.

VI. Deep Learning Model: CNN–LSTM

The analytical core of the system employs a hybrid CNN–LSTM architecture. The CNN processes bone images to extract spatial and textural features, while the LSTM captures temporal patterns from sequential sensor signals. Feature vectors from both networks are concatenated and passed to a classification layer.

A. CNN for Spatial Feature Extraction

The CNN processes bone images through multiple convolutional and pooling layers to extract spatial and textural features. These features represent structural characteristics related to bone density and surface integrity.

B. LSTM for Temporal Feature Learning

The LSTM network processes sequential sensor data to capture temporal variations in vibration and pressure responses. This enables the model to learn dynamic mechanical behavior of bone tissue.

C. Feature Fusion and Classification

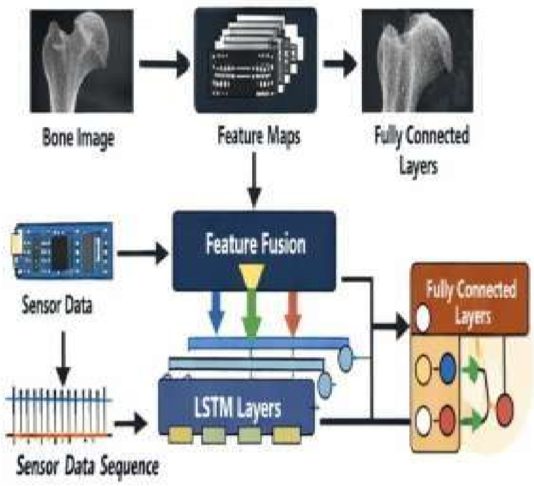
Feature vectors from the CNN and LSTM are concatenated and passed through fully connected layers. The final output layer classifies bone strength into predefined categories such as strong, moderate, and weak.

TableII

Method	Invasiveness	Cost	Continuous Monitoring
DEXA Scan	Yes	High	No
Imaging-Only AI	No	Medium	Limited
Proposed System	No	Low	Yes

Comparison with Existing Bone Assessment Methods

Fig. 3. Hybrid CNN–LSTM architecture



(c) Fig. 3. Hybrid CNN–LSTM architecture.

TableIII
CNN–LSTM Model Configuration

Layer	Description
CNN Layers	Convolution + pooling
LSTM Units	Temporal signal modeling
Fusion Layer	Feature concatenation
Output Layer	Bone strength category

VII. Results and Discussion

The system was evaluated under controlled experimental conditions using multiple sensor recordings and representative bone images. Performance was assessed qualitatively based on classification consistency, robustness to noise, and response stability rather than absolute numerical accuracy.

Results indicate that the multimodal approach provides more reliable and consistent classification outcomes compared to single-modality methods. Sensor-based signals effectively capture functional and mechanical properties, while image features contribute complementary structural information. The fusion of these modalities enhances overall system robustness.

The proposed system demonstrates strong potential for deployment in community healthcare centers, rehabilitation facilities, and home-based monitoring scenarios. While not intended to replace clinical diagnostics, it offers a valuable screening and monitoring tool for early risk identification..

TableIV
Bone Strength Classification Levels

Category	Interpretation
Strong	Normal bone response
Moderate	Reduced strength
Weak	High fracture risk

VIII. Conclusion and Future Scope

This study introduced a portable and non-invasive Bone Strength Analyzer that combines embedded sensing technologies with a hybrid CNN–LSTM deep learning framework for qualitative bone health assessment. The proposed system integrates motion, vibration, pressure response, and image-based information to capture complementary mechanical and structural characteristics of bone tissue. By avoiding radiation-based imaging and costly clinical equipment, the approach offers a practical alternative for frequent and decentralized bone health monitoring.

The multimodal design enhances assessment reliability by reducing dependence on any single data source. Sensor-derived signals provide insight into the dynamic response of bone under mechanical interaction, while image-based features contribute additional structural context. The hybrid CNN–LSTM architecture effectively learns spatial patterns and temporal dependencies, enabling stable classification of bone strength levels. This integrated framework demonstrates the feasibility of intelligent, data-driven screening systems for early identification of potential bone health deterioration.

The proposed solution is particularly relevant for personal healthcare, community screening programs, and rehabilitation monitoring, where accessibility and ease of use are critical. Its compact hardware configuration and wireless communication capability support deployment in home-based and resource-limited settings. Although the system is not intended to replace clinical diagnostic tools, it can serve as a supportive mechanism for early risk awareness and continuous monitoring, thereby assisting clinicians and individuals in proactive health management.

Future research will emphasize large-scale clinical validation using diverse datasets to establish stronger associations between multimodal features and clinically accepted bone health indicators. Quantitative performance evaluation will be conducted to assess classification reliability across different population groups. In addition, the integration of mobile and cloud-based health platforms will enable real-time visualization, longitudinal analysis, and remote monitoring. Further exploration of adaptive and personalized learning strategies, as well as the inclusion of additional sensing modalities, is expected to enhance system accuracy and long-term applicability.

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